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Session: 2025-26

LLM Fine Tuning and Optimisation



Education to Eternity

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Unit - III

Parameter Efficient Fine-Tuning (PEFT)

- **Concept, Methods and Applications**
- Parameter Efficient Fine-Tuning in Large Language Models
- Adapters, Prefix Tuning, Prompt Tuning, LoRA
- Efficient Model Customization Techniques

Concept of Parameter Efficient Fine-Tuning (PEFT)

Definition

- Parameter Efficient Fine-Tuning (PEFT) is a technique used to adapt large pre-trained language models (LLMs) to specific tasks by **updating only a small number of parameters**, while keeping most of the original model weights frozen.
- Instead of re-training the entire model, PEFT modifies or adds small task-specific components.

Introduction to PEFT

Definition

- Parameter Efficient Fine-Tuning (PEFT) is a method where only a small portion of model parameters is trained while keeping the original model frozen.
- **Main Idea**
- Reuse knowledge learned during pre-training.
- Modify only small components for new tasks.
- **Goal**
- Reduce computation and memory usage.
- Maintain performance similar to full fine-tuning.

Why PEFT is Needed in Modern AI

- LLMs contain billions of parameters.
- Training full models is expensive.
- Organizations need multiple task-specific models.
- PEFT enables:
 - Faster experimentation
 - Lower hardware requirements
 - Efficient deployment
 - Scalable AI solutions

Advantages of PEFT

- Only small parameters are trained.
- Faster training cycles.
- Reduced GPU memory usage.
- Smaller storage size for task-specific models.
- Easy switching between tasks.
- Multiple PEFT modules can share same base model.
- Suitable for edge computing and mobile devices.

PEFT Methods Overview

- Major PEFT techniques used in LLM adaptation:
- **Adapters**
 - Add small layers inside the model.
- **Prefix Tuning**
 - Add trainable prefix tokens.
- **Prompt Tuning**
 - Learn soft prompts automatically.
- **Low-Rank Adaptation (LoRA)**
 - Train low-rank weight updates.
- Each method reduces trainable parameters differently.

Adapters – Concept

- Adapters are small neural network modules inserted between transformer layers.
- Original model weights remain frozen.
- Only adapter parameters are updated during training.

Structure

- Transformer Layer → Adapter Layer → Transformer Layer
- Adapters learn task-specific transformations without modifying core model knowledge.

Adapters – Working Mechanism

- Input passes through frozen transformer layers.
- Adapter layer processes intermediate representation.
- Adapter learns task-specific information.
- Output is combined with original model output.
- **Key Idea**
- Separate general knowledge from task-specific knowledge.

Adapters – Advantages and Applications

- **Advantages**

- Modular design.
- Easy addition/removal for new tasks.
- Efficient storage.

- **Applications**

- Multi-domain NLP systems.
- Domain adaptation (medical, legal).
- Multi-task learning environments.

Prefix Tuning – Concept

- Prefix tuning adds trainable vectors before input tokens.
- These vectors guide attention layers of the model.
- No change to original model parameters.
- **Idea**
- Provide learned context that influences model behavior.

Prefix Tuning – Working

- Trainable prefix vectors are attached to input.
- Attention mechanism treats prefixes as additional context.
- Model learns how prefixes influence output generation.
- **Advantage**
- Very small number of parameters required.

Prefix Tuning – Applications

- Text generation
- Story generation
- Summarization
- Dialogue systems
- Controlled text generation

Prompt Tuning – Concept

- Prompt tuning replaces manual prompts with learned prompts.
- Soft prompts are continuous embeddings.
- Only prompt embeddings are trained.
- **Example**
- Instead of writing:
“Classify the sentiment of this text”
the model learns an embedding representing this instruction.

Prompt Tuning – Working and Advantages

- **Working**
 - Soft prompts added to input embeddings.
 - Model interprets prompts as task instructions.
- **Advantages**
 - Extremely parameter efficient.
 - Easy to scale to many tasks.
 - Requires minimal storage.

Low-Rank Adaptation (LoRA) – Concept

- Large weight matrices are expensive to update.
- LoRA decomposes weight updates into smaller low-rank matrices.
- Mathematically:
$$W = W_0 + AB$$
- Where:
- W_0 = original frozen weights
- A and B = small trainable matrices

LoRA – Working Mechanism

- Original model weights remain unchanged.
- Only low-rank matrices are trained.
- During inference, updates are combined with original weights.
- **Benefits**
- Large reduction in trainable parameters.
- Maintains high performance.

LoRA – Applications

- Chatbot customization
- Enterprise AI systems
- Domain-specific LLM training
- Code generation models
- Efficient LLM deployment

Applications of PEFT

- Domain adaptation (medical, finance, legal)
- Personalized AI assistants
- Multilingual translation systems
- Industrial AI applications
- Edge AI deployment
- Research and experimentation

Combining PEFT Methods

- **Motivation**
- Single PEFT method may not capture all requirements.
- **Examples**
- LoRA + Prompt Tuning
- Adapters + Prefix Tuning
- Hybrid PEFT frameworks
- **Benefits**
- Improved accuracy
- Flexible customization
- Efficient training

Comparison of PEFT Methods

Method	Trainable Part	Parameter Size	Best Use
Adapters	Added layers	Small	Multi-task learning
Prefix Tuning	Prefix vectors	Very small	Generation tasks
Prompt Tuning	Soft prompts	Very small	Instruction tasks
LoRA	Low-rank matrices	Small	Large LLM adaptation